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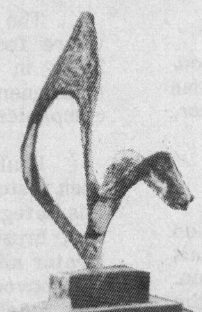

Power System Relaying Committee

AN ACCURATE FAULT LOCATOR WITH COMPENSATION FOR APPARENT REACTANCE IN THE FAULT RESISTANCE RESULTING FROM REMOTE-END INFEED

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Abstract - A microprocessor based fault locator is described, which uses novel compensation techniques to improve accuracy. It displays the distance to the fault in percent of transmission line length, for facilitating repair and restoration following a permanent fault. Also, it pinpoints weak spots following transient faults.

This new method for fault location on electric power transmission lines uses recorded phase currents and voltages at the near end. The main feature of the method is that it considers the influence of the remote-end infeed of the transmission line by using a complete network model.

A microprocessor filters the ac currents and voltages from the protective relaying instrument transformers to extract the fundamental components of the signals. It then computes the distance to the fault point, compensating for the apparent reactance in the fault resistance resulting from load current and the variations in impedance angles in the power-system network.

The design has undergone field tests and evaluation. The outcome of the field tests is presented and has confirmed the validity of the concept of the fault locator, showing an accurate display of the distance to fault.

INTRODUCTION

Distance relays for transmission-line protection provide some indication of the general area where a fault occurred, but they are not designed to pinpoint the location. Since power circuit breakers are only installed at the terminals, it is immaterial where the line faulted for isolating the flashover. However, with immediate knowledge of the location, the nature (type and measuring data) of the fault can be determined quickly, facilitating repair and restoration. A locator is also useful for transient faults, pointing to a weak spot that is threatening further trouble.

Various methods were developed during the recent years to detect the location of fault on a transmission line. They are mostly based on analog techniques[2]. Some fault locator designs are in service which are reliable in detecting permanent faults. New methods and systems of fault location based on both analog and digital techniques seem to offer good prospects to obtain a precise location of transient faults also. The method reported in [1] is an approximate method. The method of fault location technique described here seems more advantageous, since it takes into consideration the effects of both ends of the line.

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This novel approach is accomplished by using a complete network model, where the infeed from the network beyond the remote end point is rigorously taken into consideration.

Pre-fault load-current samples are stored and used for compensation to eliminate a substantial effect on accuracy. Also, representative values for the source impedances are stored to compensate for variations in impedance angles. This novel approach is described as well as the system design and performance.

PURPOSE FOR FAULT LOCATORS

Even where helicopters are immediately available for patrol following unsuccessful reclosing, fault locators perform a valuable service. Trouble cannot always be found with a routine patrol with no indication of where the fault occurred. For example, tree growth could reduce clearances, resulting in a flashover during severe conductor sagging. By the time the patrol arrives, the conductors have cooled, increasing the clearance to the tree. The weak spot is not obvious.

The importance of fault locators is more obvious where foot patrols are relied upon, particularly on long lines, in rough terrain. Also, locators can help where maintenance jurisdiction is divided between different companies or divisions within a company.

Fault locators are valuable even where the line has been restored either automatically or non-automatically. In this category are faults caused by cranes swinging into the line, brushfires, damaged insulators and vandalism. The locator allows rapid arrival at the site before the evidence is removed or the "trail becomes cold". Also, the knowledge that repeat faults are occurring in the same area can be valuable in detecting the cause. Weak spots that are not obvious may be found because a more thorough inspection can be focussed in the limited area defined by the fault locator.

FAULT LOCATION FUNDAMENTALS

The fault location computations determine the apparent fault impedance with novel compensation for the fault resistance drop, eliminating the errors inherent in conventional reactance-type measurements. For a fault-protection relay these errors are tolerable because a safety margin is inserted in its setting. However, for the fault locator a more precise measurement is quite desirable. The Appendix derives the equation for the apparent impedance seen by a reactance-type measurement. Figs. A1 and A2 illustrate the apparent reactance effect in the fault-resistance term.

Fig. 1 shows the connection of the fault locator at station A. Using the ac quantities available at station A, it is not possible to determine the total fault current I_F unless p and Z_{SB} are known; conventional devices tolerate the errors resulting from the infeed current I_B flowing through the fault resistance R_F, out of phase with respect to I_A.

In the method reported in Reference 1, the current distribution factors for the parts of the network, located on either side of the fault point, are assumed to have the same arguments, or else it is assumed that the difference between the arguments is known and constant along the line segment

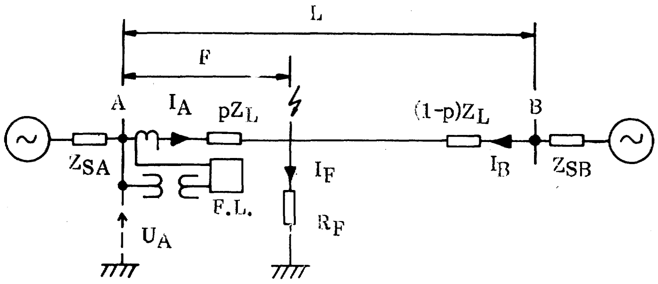


Fig. 1 Power system one-line diagram with fault.

in question. This simplified assumption can give rise to substantial errors in the calculated fault distance. The difference between the two arguments mentioned generally varies with the distance to the fault.

The fault locator program described here utilizes representative values of the source impedances to determine a correct description of the network. The value of R_F is unknown; however, it is not needed — only the angle of $I_F R_F$, the fault point voltage, is used.

The other key aspect of this locator's algorithm is its use of pre-fault current memory to determine the change in line current caused by the fault: the actual fault current minus the prefault value. Eq. (A9) from the appendix, repeated here as Eq. (1) states that the station A voltage is the sum of the drops in the line to the fault point plus the fault point voltage:

$$U_{1A} = I_{1A}\, p\, Z_{1L} + I_{1F} R_F \tag{1}$$

Eq. (1) is written for a 3-phase fault, using the positive sequence notations. Eq. (2) expresses the same concept in general terms applicable to any fault:

$$U_A = I_A\, p\, Z_L + I_F R_F \tag{2}$$

To consider load current effects, the actual voltages and currents can be considered to be composed of the prefault values plus the changes caused by the fault. The appendix discusses this concept. The load currents are generated in Fig. 2 by the angular difference between the source voltages. Voltage U_F exists at the fault point prior to the fault and is the driving voltage in Fig. 3 producing the changes caused by the fault.

From Fig. 3, the change in positive-sequence current in the line at A, ΔI_{1A}, is related to the total positive-sequence current by the current distribution factor D_{1A}:

$$\Delta I_{1A} = D_{1A}\, I_{1F} \tag{3}$$

Writing a similar, but general expression:

$$I_{FA} = D_A I_F \tag{4}$$

where I_{FA} is the current change produced by the fault, equal to the actual fault current less the prefault current. The expression for I_{FA} varies with the fault type; Table I defines these. Changes in phase currents are used except for single-line-ground faults. The latter use the change in faulted phase current less the zero sequence current I_{0A}. The zero-sequence current is extracted because the zero-sequence distribution factor D_{0A} is not known as reliably as the positive-sequence factor D_{1A}. The 3/2 factor in Table I provides heavier weighting to compensate for the removal of the zero-sequence current.

The I_{FA} expression for a single-line-ground (SLG) fault will now be derived with the objective of eliminating the

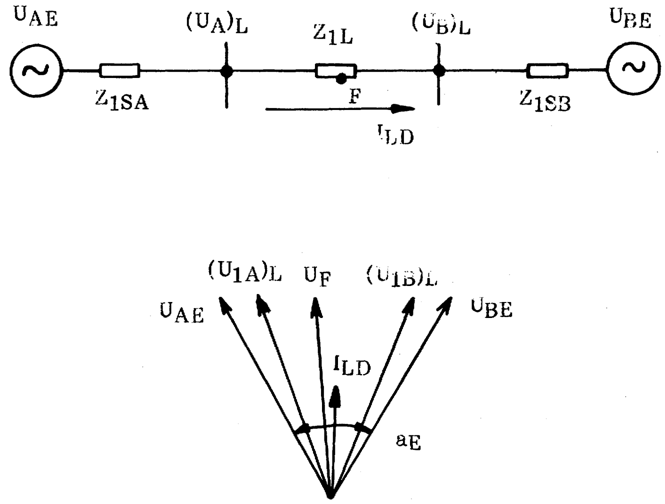


Fig. 2 Prefault conditions.

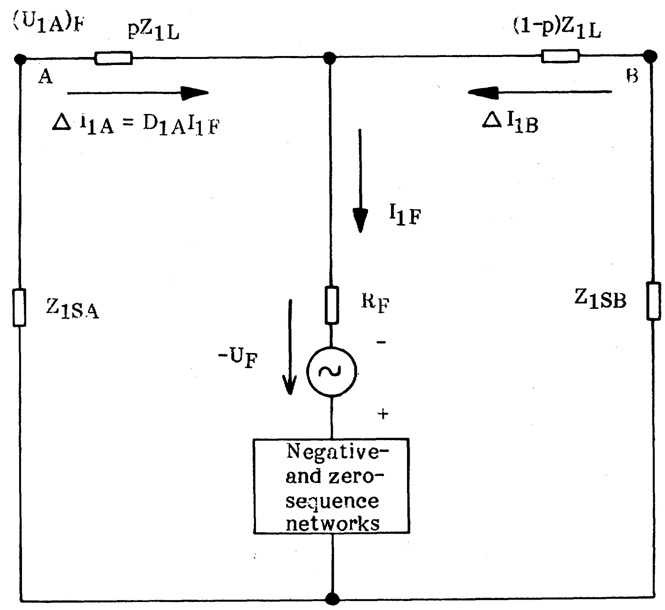


Fig. 3 Faulted network (Thevenin equivalent).

processing of the zero-sequence components to obtain I_F. Fig. 4 shows the current relations in the fault for an SLG fault:

$$\frac{I_F}{3} = I_{1F} = I_{2F} = I_{0F} \tag{5}$$

Eliminating the zero-sequence current I_{0F} :

$$I_F = 3/2\ (I_{1F} + I_{2F})$$
$$= 3/2\ \left(\frac{\Delta I_{1A} + \Delta I_{2A}}{D_{1A}}\right) \tag{6}$$

The distribution factors for the positive- and negative-sequence currents may be assumed, with great accuracy, to be equal (i.e., $D_{1A} = D_{2A}$).

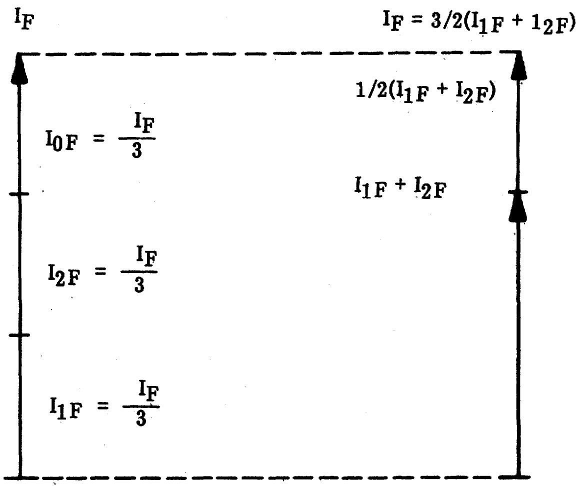


Fig. 4 Current relations for a single-line-to-ground fault.

At station A in the line:

$$\Delta I_A = \Delta I_{1A} + \Delta I_{2A} + \Delta I_{0A} \tag{7}$$

or

$$\Delta I_{1A} + \Delta I_{2A} = \Delta I_A - \Delta I_{0A} \tag{8}$$

Neglecting pre-fault zero-sequence current:

$$\Delta I_{1A} + \Delta I_{2A} = \Delta I_A - I_{0A} \tag{9}$$

Eq. (9) in (6) :

$$I_F = 3/2 \; \frac{\Delta I_A - I_{0A}}{D_{1A}} \tag{10}$$

Let $I_{FA} = 3/2\,(\Delta I_A - I_{0A})$ (11)

The general expression of Eq. (2) can be rewritten as:

$$U_A = I_A\, p\, Z_L + \left(\frac{I_{FA}}{D_A}\right) R_F \tag{12}$$

From Fig. 1 :

$$D_A = \frac{(1-p)\,Z_L + Z_{SB}}{Z_{SA} + Z_L + Z_{SB}} \tag{13}$$

Substituting Eq. (13) in (12) and rearranging, yields:

$$p^2 - p\,K_1 + K_2 - K_3\,R_F = 0 \tag{14}$$

where:

$$K_1 = \frac{U_A}{I_A Z_L} + 1 + \frac{Z_{SB}}{Z_L} \tag{15}$$

$$K_2 = \frac{U_A}{I_A Z_L}\left(\frac{Z_{SB}}{Z_L} + 1\right) \tag{16}$$

$$K_3 = \frac{I_{FA}}{I_A Z_L}\left(\frac{Z_{SA} + Z_{SB}}{Z_L} + 1\right) \tag{17}$$

The complex expression of Eq. (14) contains the unknowns p and R_F. However, Eq. (14) can be separated into two simultaneous equations, one real and one imaginary. By eliminating R_F, a single expression results with the single unknown p. This is solved by the program, using the peak values and their phase position, taken from the Fourier analysis routine which yields the fundamental components of the signals.

Fig. 5 provides an alternative view of how the fault locator determines p without requiring the scalar value of the fault-point voltage. Fig. 5 is a triangulation problem to determine the intersection point F, knowing U_A and I_A by measurement and computing a_D and a_L. While the computer is not specifically using this algorithm, nevertheless the conclusions reached using the concept of Fig. 5 are valid: viz. that the magnitude of $I_F R_F$ is not required, but only its angle a_D.

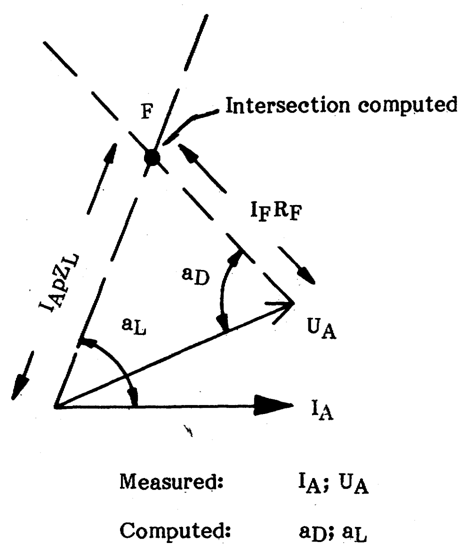


Fig. 5 Fault-resistance-drop compensation.

Parallel Line Cases

The locator can be finished with a modified algotithm for a parallel-line application. The positive-sequence network is completely described by re-defining equation 13 as:

$$D_A = \frac{(1-p)\,(Z_{SA} + Z_{SB} + Z_L) + Z_{SB}}{2\,Z_{SA} + 2\,Z_{SB} + Z_L} \tag{13A}$$

Eq. (13A) assumes an identical parallel line. Use of the equation will improve accuracy with the parallel line in service.

Zero-sequence mutual coupling can be compensated for by interconnecting two fault locators . With a fault locator on the parallel line its residual current reading can be input to its companion locator via the local printer loop circuit; the mutual resistance and reactance are input as additional setting parameters.

ALGORITHMS

The fault location algorithm uses the pre-fault and fault currents and voltages at the near end of the transmission line to determine the distance to fault. Current and voltage samples are continuously measured. Following a signal from the line protection at the instant it initiates breaker tripping, current and voltage samples for six cycles are frozen until the completion of the distance-to-fault computation.

TABLE I QUANTITIES PROCESSED FOR VARIOUS FAULT TYPES

Type of Fault	U_A	I_A	I_{FA}
RN	U_{RA}	$I_{RA} + K_N \times I_{NA}$	$3/2\ (\Delta I_{RA} - I_{0A})$
SN	U_{SA}	$I_{SA} + K_N \times I_{NA}$	$3/2\ (\Delta I_{SA} - I_{0A})$
TN	U_{TA}	$I_{TA} + K_N \times I_{NA}$	$3/2\ (\Delta I_{TA} - I_{0A})$
RST RS RSN	$U_{RA} - U_{SA}$	$I_{RA} - I_{SA}$	ΔI_{RSA}
ST STN	$U_{SA} - U_{TA}$	$I_{SA} - I_{TA}$	ΔI_{STA}
TR TRN	$U_{TA} - U_{RA}$	$I_{TA} - I_{RA}$	ΔI_{TRA}

The program then determines the loop of analog data (currents and voltages) on which to base the computation. For example, the RS loop is selected if the R and S phase selectors operate. Table I shows the normal loop selection. For double-line-ground faults only, a phase-to-ground loop may be selected instead of the normal phase-phase loop, if desired. Table II shows the loop options. The program next determines the fault inception point within the 6 cycle data, looking at the selected loop ac quantities, choosing the current(s) first. If necessary, the voltage(s) is also processed to find the fault inception point.

Reading 24 samples apart are compared for a significant change, starting with oldest samples. The required threshold for the current change is adaptive, depending on the prefault current level. The voltage threshold is fixed. If no change is found, a one sample advance occurs and the procedure is repeated.

TABLE II LOOP SELECTION FOR DOUBLE-LINE-TO-GROUND FAULTS

Phase Input	Loop Selected: Normal	Cyclic	Acyclic
RSN	RS	SN	RN
STN	ST	TN	TN
TRN	TR	RN	RN

If unable to find the fault-inception point, the program reverts to a "slow start" algorithm, which eliminates the load-current compensation. This would occur for a switch-into-the-fault case or a time delay trip for an end-zone fault, where in such cases no prefault samples are available. In both cases, there would be no load flow to compensate, because at the time of tripping the far end of the line would be open.

When a two-phase loop is selected (e.g. R and S for an RSN selector input), two different fault inception points will generally occur; the program selects the latter of the two fault points.

The program selects 24 samples from the two periods spanned by SL1 to SF1 in Fig. 6, where SF1 is the third sample following the fault point. For the fault data, 24 samples immediately following SF1 in Fig. 6 are selected.

The pre-fault and fault currents and voltages on all phases are filtered by Fourier analysis to yield the fundamental component. The selected loop quantities are then processed for the distance-to-fault determination, while the complete set is available for printing for user analysis.

The fundamental components are determined by multiplying each sample by the appropriate instantaneous sine value and integrating over a full period, then repeating the process by multiplying by cosine values. The result provides the scalar value and argument for each ac quantity. The Fourier filtering effectively attenuates the dc offset component and power system harmonics plus CCVT transients and CT saturation distortion.

The program uses the peak and angle outputs from the filtering routine to compute the distance to the fault.

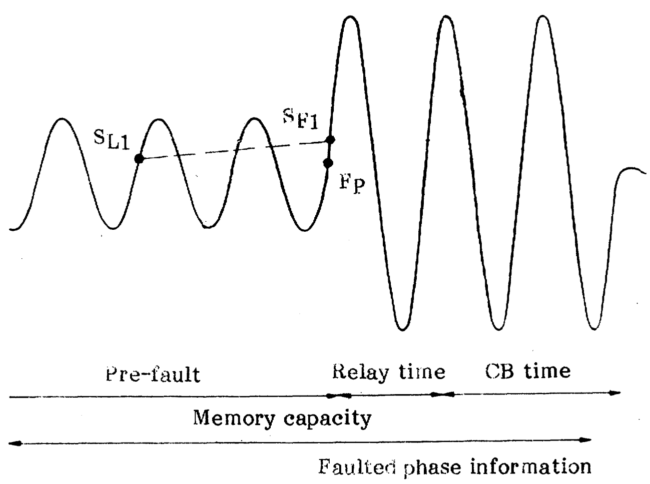


Fig. 6 Samples are selected from two periods before and after fault inception point F_P.

HARDWARE

Fig. 7 shows the main elements of the hardware. The digital inputs consist of the start signal from the line protection relay breaker-trip output and the phases R, S, T (i.e., A, B, C) and ground phase selection outputs from the line protection or from the integral phase underimpedance and ground overcurrent phase-selector units. The start input initiates a distance-to-fault computation.

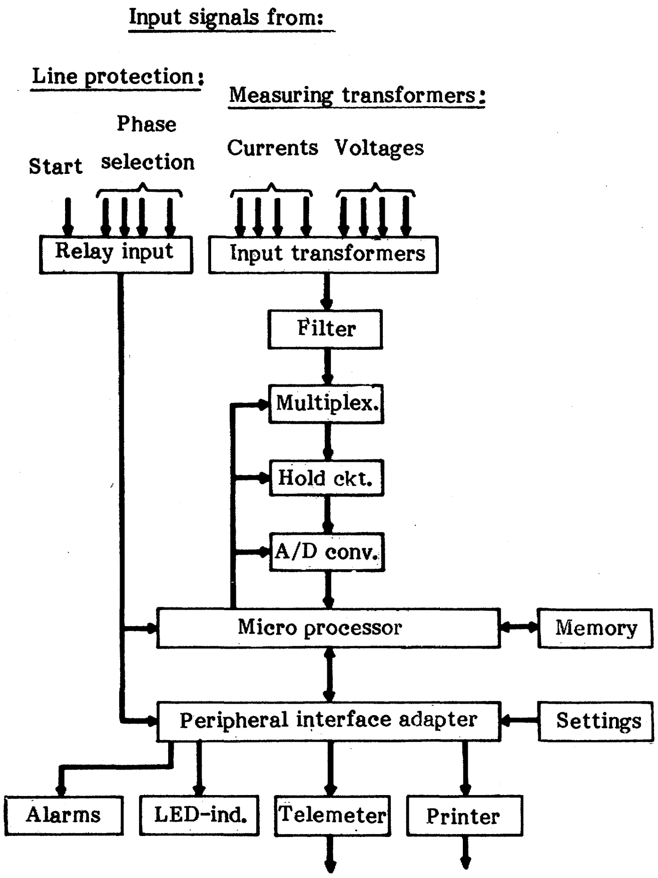


Fig. 7 Hardware configuration.

The 3-phase currents and voltages enter through input transformers which provide galvanic isolation from the instrument transformers, as well as transforming the signals to a level suitable to the electronics. A screen between the windings minimizes common-mode surge coupling. The analog signals feed through low-pass filter for signal conditioning, using a 500 Hz cut-off frequency. The filter outputs are switched in sequence by the multiplexer and fed into the hold circuit in preparation for conversion to a digital value proportional to the instantaneous value of the ac wave. Both the digitized signals and the relay input status are stored in a 6-cycle circular file in the memory with the aid of the microprocessor (MC 6803 microprocessor, 8 bit, 814 ns memory cycle).

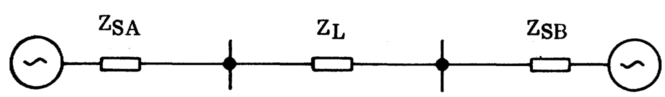
The microprocessor processes the measuring values according to the fault location algorithm and presents the distance to fault in percentage of the line length on the LED indicator or remote connected equipment (printer, telemeter, etc.). The microprocessor continuously executes a monitoring routine. If the computer fails to periodically output an "all's well" signal, a peripheral circuit operates an alarm relay. During normal service, a pushbutton is used for initiating test programs to the memory and for resetting the LED indicator.

Table III shows the parameters to be set by the user. See GLOSSARY OF SYMBOLS. Parameters 1 to 4 are the line constants; 5 to 8 are the source constants. The source impedances will change with system conditions, so a representative value is selected. The inclusion of the source impedances in the fault location algorithm is a novel approach, which allows compensation for different impedance angles in the power system network.

The parameters are set with the help of thumbwheels. A pushbutton operation initiates an EPROM (electrically programmable read-only memory, 22 Kbytes used) burn-in. The digits of parameter No. 9 determine four different items listed in Table III.

TABLE III: SETTING & PROGRAMMING PARAMETERS

1.	R_{1L}	7.	R_{1SB}
2.	X_{1L}	8.	X_{1SB}
3.	R_{0L}	9.	- Type of phase selection (Normal, cyclic, acyclic)
4.	X_{0L}		- CT polarity
			- Printout (A,B,C)
5.	R_{1SA}		- Line number
6.	X_{1SA}		



DESIGN

Fig. 8 shows the fault locator assembly with phase selector and printer, which is suitable for 19" rack mounting. The assembly consists of test unit, power supply unit, transformer unit and measuring shunt unit, which are directly mounted to and held together by two apparatus bars. The remaining plug-in units are connected to a mother-board, fed from the shunt unit. The bottom half of the assembly consists of the fault-type selection underimpedance and ground overcurrent units and the printer.

The result of the distance-to-fault calculation is shown on the LED indicator. Fig. 9 shows an example of output results from the printer. The relative distance to fault is expressed in percentage of the line length. "Phase = RN" indicates A and ground inputs from the phase selectors (i.e., a phase R to N fault). "LOOP = RN" indicates that the computer processed phase A to ground voltage and phase A and residual currents to determine the fault location. The computed r.m.s. values of the fundamental component of ac signals in polar form prior to and during the fault are also shown in Fig. 9.

DESIGN TESTS

Model-line tests were performed on six fault locator units at the factory under dynamic conditions. A 100 km, 170 kV line was modelled using CT's and PT's or a CCVT model. The line impedances referred to 380 V were: Z_{1L} = 2.2 + j 17.0 and Z_{0L} = 16.4 + j 68.0 ohms. The ohmic values

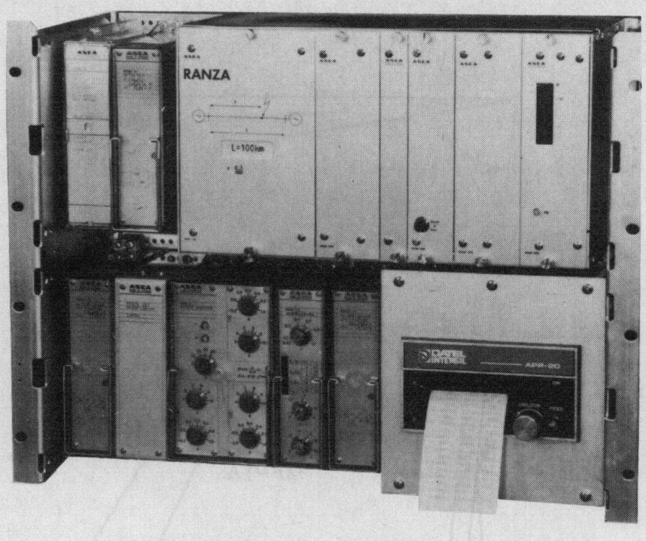


Fig. 8 Fault locator assembly with phase selector and printer.

stated below are also referred to 380 V, the primary of the system simulator. Faults were applied with varying incidence angles. Three series of tests were conducted.

Series No. 1 tests were run with zero fault resistance and rated load current, exported power from station A; the errors were less than 2 % of the line setting. Series No. 2 were run with 50 % load, with fault resistance variations from 5 to 40 ohms; the errors were less than 4 % of the line setting. Series No. 3 tests checked for errors resulting from variations in source impedance from set values; errors were less than 2 % of the line setting. Source A was varied from 2 to 32 ohms, B from 6 to 23 ohms. All ohmic values are on a 1 A rated base.

The locator was also run through the normal test series applied to protective relays, including disturbance (surge) tests: a 4-8 kV fast transient test per SEN361503 (a fast rise test simulating inductive interruptions in the control circuits) and the IEC 255-4 1 MHz, 2.5 kV test.

FIELD RESULTS

1) Two prototype units were installed for evaluation in Swedish utilities. No misoperations or component failures have been experienced. One of the prototypes was installed on the 30th of June 1982, on a 76 km, 130 kV transmission line between Nybro and Vaxjo, on the Sydkraft transmission system in southern Sweden.

On July 2nd 1982 , a disturbance was detected on the line between Nybro and Vaxjo. The fault locator display showed 89 % (corresponding to 67.6 km). The type of fault that was printed out was a single-phase ground fault on phase T. An aerial inspection of the transmission line was made on July 6th 1982 and the point of disturbance was found in phase T at a distance of 67.0 km from Nybro. The flashover occurred from conductor to arcing horns, across a tower insulator string. The distance to fault, calculated by the fault locator deviated from the actual distance to fault by less than 1 %. Load current was 238A and the calculated fault resistance was 7.2 ohms. The parameter settings matched those existing in the network at the time of the fault.

The other prototype was installed on a 400 kV Swedish State Power transmission line. However, no fault has occurred on that line.

Line number 1

Relative distance to
fault p = 75%

Phase = RN, Loop = RN

IR	=	AMPL.	=	002,213 A	
		ARG.	=	028,4 DEG.	
IS	=	AMPL.	=	000,292 A	
		ARG.	=	308,7 DEG	
IT	=	AMPL.	=	000,831 A	
		ARG.	=	212,0 DEG	
IRØ	=	AMPL.	=	000,519 A	Pre-fault
		ARG.	=	103,1 DEG	
ISØ	=	AMPL.	=	000,514 A	
		ARG.	=	343,3 DEG.	
ITØ	=	AMPL.	=	000,524 A	
		ARG.	=	223,1 DEG.	
IN	=	AMPL.	=	001,475 A	
		ARG.	=	015,1 DEG.	
UR	=	AMPL.	=	052,157 V	
		ARG.	=	093,0 DEG.	
US	=	AMPL.	=	062,611 V	
		ARG.	=	332,9 DEG.	
UT	=	AMPL.	=	063,192 V	
		ARG.	=	210,5 DEG.	
URØ	=	AMPL.	=	063,510 V	Pre-fault
		ARG.	=	092,0 DEG.	
USØ	=	AMPL.	=	063,672 V	
		ARG.	=	331,7 DEG.	
UTØ	=	AMPL.	=	063,774 V	
		ARG.	=	211,9 DEG.	

Fig. 9 Printer output.

2) A staged fault test was conducted jointly by Texas Power & Light Co. and Texas -New Mexico Power Co. on the described fault locator. A fault locator was installed at Northwest Carrollton sub as shown in Fig. 10.

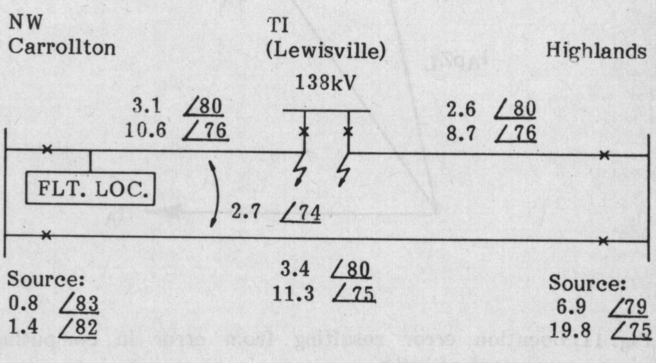


Fig. 10 Staged faults were applied at TI (Lewisville).

There is mutual coupling over 59 % of the Northwest Carrollton - TI (Lewisville) line with the Northwest Carrollton - Highlands line. It was the intention to estimate the performance of the fault locator on that line without any compensation for the mutual impedance.

Five faults, two phase-to-phase and three phase-to-ground faults were placed at the Lewisville substation during the early morning of October 25th 1983. The faults were applied at the TI (Lewisville) substation where both lines loop into the substation. The actual distance to fault was 55 % of the line. The load current was about 100A. Estimated fault resistance was 1.6 ohms for the ground faults.

The locator functioned properly for all five faults. The highest deviation of the results between the distance to fault shown on the fault locator display and the actual distance to fault was 3 %.

ACCURACY ANALYSIS

Results of overall accuracy during dynamic model-line tests are described under DESIGN TESTS. This section will discuss the accuracy of this locator in comparison with those using other approaches.

The fault locator described above computes the distance to the fault point by compensating for the apparent reactance in the fault resistance resulting from load current and variations in the impedance angles in the power system network. However, minor errors (see Fig. 11) can occur if the settings are not adequate. These errors should be moderate for practical situations. For one, the value of the fault locators increases with line length; for these applications the line impedance tends to mask the effect of source impedance variations. Again for longer lines, the fault resistance, particularly for single-phase-ground faults, becomes a smaller percent of the line impedance.

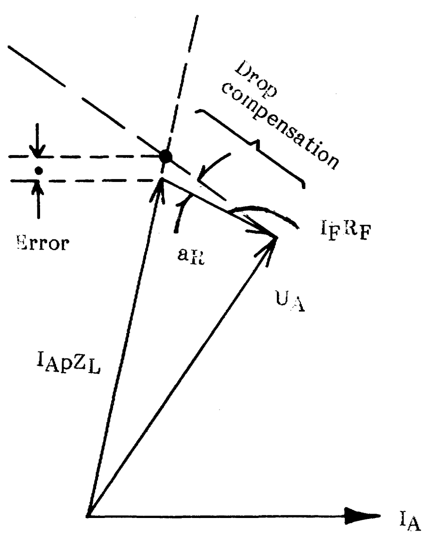


Fig. 11: Location error resulting from error in computing argument of $I_F R_F$.

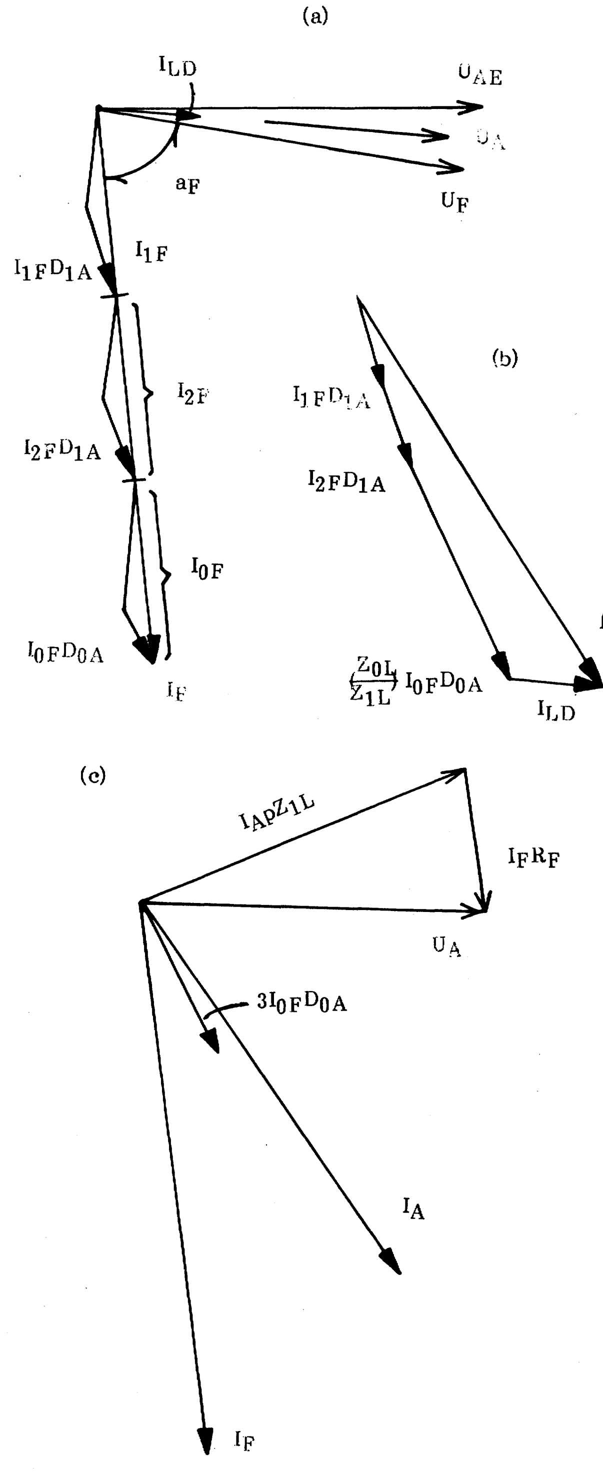


Fig. 12: System quantities for a single-line-ground fault with load export from station A.

The next series of illustrations shows qualitatively the fault-resistance infeed effects on two alternative methods of determining fault location: the reactance and the zero-sequence current-reference techniques. Two examples are shown: one for load export; the second, import.

Fig. 12(a) shows the station A line currents out of phase with the total fault current I_F. Stating it another way, source B produces out-of-phase infeeds in the fault resistance. The total phase current at A is developed in Fig. 12(b) and is the sum of the currents resulting from the fault and driven by U_F, and the pre-fault current I_{LD}. Fig. 12(c) extracts from Figs. 12(a) and (b) the essentials for developing the error diagrams of Figs. 13 and 14.

The reactance type of Fig. 13 uses the same basis of measurement as does the conventional reactance relay. The angular shift of $I_F R_F$ produces an error which can become

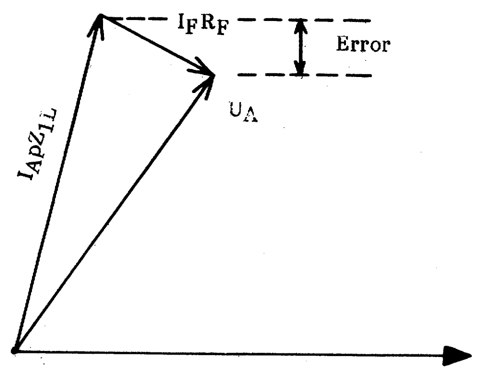


Fig. 13: Reactance-type error resulting from conditions in Fig. 12.

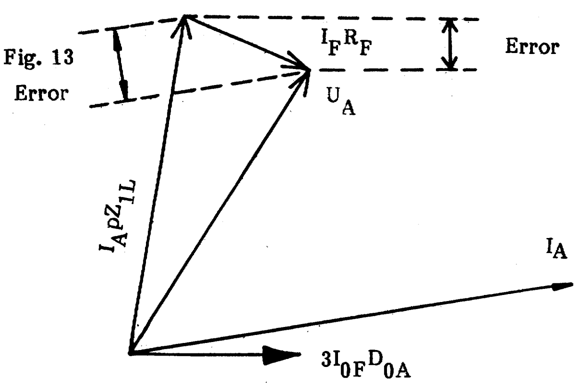


Fig. 14 Zero-sequence-current-reference-type error resulting from conditions in Fig. 12.

significant for the fault location function where greater precision is needed than for a protective relay.

The type shown in Fig. 14 differs from the reactance type in its use of residual current, $3I_{0F}D_{0A} = 3I_{0A}$ as the reference rather than using I_A. For the example of Fig. 12, the two currents $3I_{0F}D_{0A}$ and I_A are nearly in the same phase position, so the error in Fig. 14 is almost as much as the Fig. 13 error.

In Fig. 15(a), power is being imported to Station A so I_{LD} is about 180° from its position in Fig. 12. The fault currents here are identical to those of Fig. 12; however, the superposition of load current results in a different magnitude and angle for I_A. Compare 16 and 17: the latter now has a larger error because its reference current, $3I_{0F}D_{0A}$, leads the I_A position.

The load compensation method [1] will experience errors generally less than those shown in Figs. 14 and 17, depending upon the amount of out-of-phase infeed into the fault from the far end. This can be seen in the following numerical examples.

<u>Single-Line Examples</u>

Table IV shows the location errors for three cases, comparing four computational methods, using an off-line computer simulatation. Table V defines the power system conditions for two different transmission lines. These were

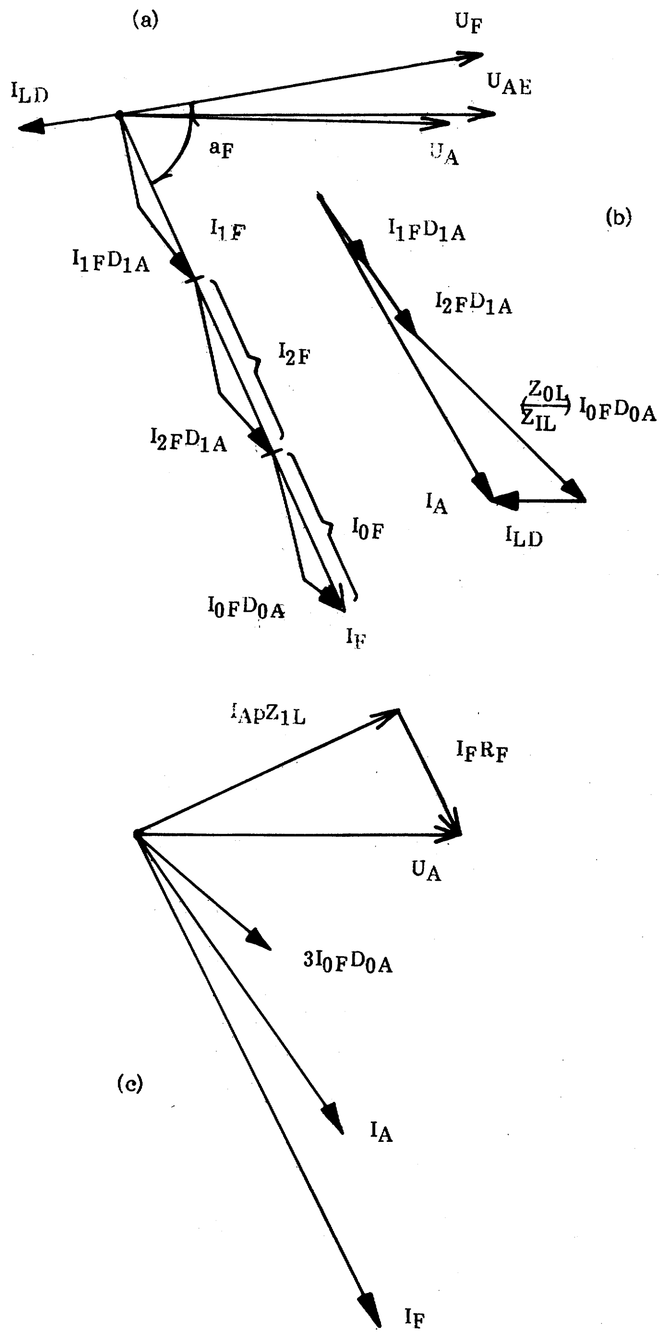


Fig. 15: System quantities for a single-phase-ground fault with load import into station A.

then faulted as specified in Table IV. Method 1 has zero error because the source impedance values were input, describing the complete system. Method 4 shows substantial errors, although smaller than both methods 2 and 3.

Case 3 in Table IV yields greater errors than cases 1 and 2 because the zero-sequence line impedance argument of 76.7 degrees is lower with respect to the zero-sequence source impedance argument. The corresponding angle for cases 1 and 2 is 81.7 degrees. Thus, the zero-sequence out-of-phase infeed is much greater in case 3, causing greater errors for methods 2, 3 and 4.

With smaller fault resistances the errors in Table IV will be correspondingly smaller. On the other hand, lower voltage lines with lower arguments will produce larger

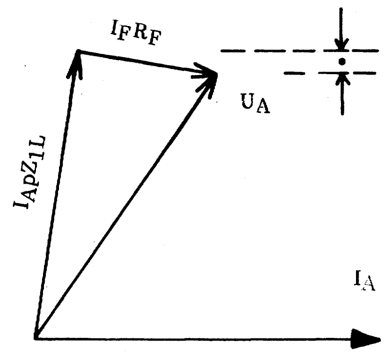


Fig. 16: Reactance-type error resulting from conditions in Fig. 15.

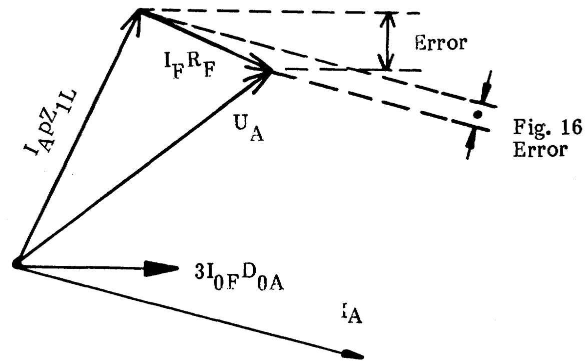


Fig. 17 Zero-sequence-current-reference-type error resulting from conditions in Fig. 15.

TABLE IV COMPARISON OF FAULT LOCATOR ERRORS FOR THE APPLICATIONS IN TABLE V

					CALCULATED RESULTS							
					Method 1		Method 2		Method 3		Method 4	
Case	Power System	Fault Locator At:	Fault Point	R_F Ohms	p	Error	p	Error	p	Error	p	Error
1	Ashe-Marion	Ashe	0.9	50	0.9	0	0.87	-3.1	0.62	-28.4	0.88	-1.9
2	Ashe-Marion	Ashe	0.9	100	0.9	0	0.86	-4.5	0.54	-36.7	0.87	-2.7
3	Stenkullen-Borgvik	Sten-kullen	0.8	125	0.8	0	0.72	-7.9	0.39	-41.0	0.76	-4.2

Method 1 As described in this paper.

Method 2 Zero-sequence-current-reference type (see Fig. 14) K_N as defined in the Glossary

Method 3 Reactance type (see Fig. 13) K_N as defined in the Glossary.

Method 4 Load compensated [1] .

Errors are in percentage of total length of line.

errors where the remote source argument is higher, such as would be the case near a generating station.

Parallel-Line Examples

Table VI shows computer-simulation results for two parallel-line cases. In both instances the parallel line is identical to its companion listed in Table V. The errors in Table VI for method 1 would occur if parallel-line compensation were not included. With both the positive - and zero-sequence effects included in the algorithms, the computed errors were zero. The uncompensated errors for the Lieto-Forsa application are greater because of the lower line arguments compared to the Ashe-Marion lines. The parallel-line effect upon accuracy will be the greatest for far-end faults, because these produce the maximum parallel-line current.

APPLICATION CONSIDERATIONS

The unit is suitable for use with lines or sources with secondary impedances in the range of 0-1000 ohms based on a 1 A CT (0-200 ohms for 5 A CT's). Signal current range is 0.1-20 times rated; voltage, 0.01-1.5 rated. It is available with 1, 2 and 5 A ratings, withstanding 3 times rated current continuously.

It is designed for use with the relaying instrument transformers and introduces a burden of 1 VA per phase at rated voltage or current.

The unit is designed and tested just as if it were a protective relay.

Operating temperature range is 0 to 55° C.

It requires 1.5 cycles of fault duration for an accurate measurement.

It is not necessary to apply one locator at each line terminal. Four units can share one optional printer.

The distance to the fault can be remotely logged using the built-in telemetering outputs from the locator.

TABLE V SYSTEM DESCRIPTIONS FOR EXAMPLES IN TABLES IV AND VI

		Ashe-Marion (USA)	Stenkullen-Borgvik (SWEDEN)	Lieto-Forsa (FINLAND)
Voltage - kV		525	400	110
Z_{1SA}	mag.	17.6	21.3	5.8
	arg.	90	90	85
Z_{0SA}	mag.	6.2	24.4	70
	arg.	90	90	85
Z_{1SB}	mag.	42.2	25.6	15
	arg.	90	90	85
Z_{0SB}	mag.	29.5	27.8	130
	arg.	90	90	80
Z_{1L}	mag.	118.5	53.5	27.6
	arg.	87.0	86.4	73.7
Z_{0L}	mag.	463.0	247.6	108.7
	arg.	81.7	76.7	74.3
U_{AE}	mag.	320.4	225.2	63.5
	arg.	-17.0	0	0
U_{BE}	mag.	334.9	240.2	63.5
	arg.	-67.0	-29.4	-15
I_{LD}	mag.	1555	1184	
	arg.	-37.0	-5.7	

Impedances in ohms
Voltages in kV
Currents in A
Arguments in degrees

SUMMARY

1 The fault locator displays the percent distance to the fault, using novel compensation techniques for the errors resulting from remote-end source infeed into the fault resistance. This is accomplished by using a complete network model.

2 Compensation for fault-resistance voltage drop utilizes prefault current and representative values for source resistance and reactance. The inclusion of source impedances is a novel approach which provides improved accurancy.

3 The microprocessor-based system is designed and built to protective-relaying standards, including all surge withstand requirements.

4 It is suitable for use with the relaying instrument transformers, introducing negligible burden.

5 Extensive software filtering minimizes the effects of power-system and instrument-transformer transients.

6 The microprocessor monitors itself and built-in functional-test facilities can pinpoint a particular chip or analog-system failure.

7 The locator can function either with the line-relay phase selectors or with the optional built-in relay units.

8 Only one end of the line needs to be equipped.

TABLE VI FAULT LOCATOR ERRORS (METHOD 1) FOR PARALLEL LINE APPLICATIONS WITHOUT COMPENSATION

Fault locator installed at Ashe and Lieto.

		Fault Location	R_F-Ohms	Ashe Marion	Lieto Forsa
Z_{0M}	mag	-	-	332.1	75.8
	arg.	-	-	73.6	73.5
I_{LD}	mag	-	-	1165	240
	arg.	-	-	-37.8	4.3
Percent Error:					
		0.3	50	0.1	
		0.3	100	0.1	
		0.4	4.3		0.4
		0.4	20		0.4
		0.8	50	3.2	
		0.8	100	2.9	
		0.8	4.3		15.8

REFERENCES

1) T. Takagi, Y. Yamakoshi. M. Yamaura, R. Kondow, T. Matsushima: "Development of a new type fault locator using the one-terminal voltage and current data", IEEE Trans., VOL. PAS-101, No. 8, August 1982, pp 2892-2898.

2) M. Souillard, Ph. Sarquiz, L. Mouton: "Development of measurement principles and of the technology of protection systems and fault location systems for three-phase transmission lines", CIGRE No. 34-02, August 1974.

GLOSSARY OF SYMBOLS

a_D angle of fault resistance drop, $I_F R_F$, with reference to U_A

a_E angle between generated voltages U_{AE} and U_{BE}

a_F angle of total fault current, with reference to U_F

a_L argument of line impedance

a_R angular error in computing $I_F R_F$

D_A general expression for current distribution factor at station A.

D_{1A}, D_{2A}, D_{0A} positive-, negative- and zero-sequence current distribution factor in line at station A

I_A, I_B line current on faulted phase(s) at station A and B, respectively. General, i.e. not referring to a specific symmetrical component sequence

I_{1A} total positive-sequence current in line at station A.

ΔI_{1A}, ΔI_{1B} positive-sequence current change as a result of fault, in line at station A and B, respectively

I_F total fault current

I_{FA} change in line current at station A, as a result of fault. General - see Table I for specific definition depending upon fault type

I_{1F}, I_{2F}, I_{0F} positive-, negative- and zero-sequence currents in fault

I_{LD} load current in line

I_{NA} residual current ($3\ I_{0A}$) in the line at station A

I_{0A} zero-sequence line current at station A

I_{RA}, I_{SA}, I_{TA} current in line at station A, phases R, S and T, respectively

ΔI_{RA}, etc.—total current minus pre-fault current, in the line at station A, phase R

K_N $\dfrac{Z_{0L} - Z_{1L}}{3Z_{1L}}$

N ground, when referring to fault type

p proportion of the total line from station A (fault locator location) to the fault

R, S, T power system phases (alternative: A,B,C)

R_{1L}, R_{0L} positive-and zero-sequence line resistances

R_{1SA}, R_{1SB} positive-sequence source resistance at station A and B, respectively

R_F fault resistance

U_A, U_B voltage on faulted phase(s) at station A and B, respectively. General, i.e. not referring to a specific symmetrical component

U_{AE}, U_{BE} source voltage behind equivalent impedance at station A and B, respectively

U_{1A} actual positive-sequence voltage at station A

$(U_{1A})_F$ positive-sequence change in voltage at station A resulting from fault

$(U_{1A})_L$, $(U_{1B})_L$ positive-sequence voltage prior to fault at station A and B, respectively

U_F voltage at fault point prior to fault

U_{RA}, U_{SA}, U_{TA} voltage to ground at station A, phases R, S and T, respectively

X_{1L}, X_{0L} positive-and zero-sequence line reactance

X_{1SA}, X_{1SB} positive-sequence source reactance at station A and B, respectively

Z_A apparent impedance on the line at station A

Z_{1SA}, Z_{1SB} positive-sequence source impedance at station A and B, respectively

Z_{0SA}, Z_{0SB} zero-sequence source impedance at station A and B, respectively

Z_{1L}, Z_{0L} positive- and zero-sequence line impedance

Z_{SA}, Z_{SB} source impedance behind station A and B, respectively. General, i.e. not referring to a specific symmetrical component sequence

Z_L total line impedance. General, i.e. not referring to a specific symmetrical component sequence

Z_{0M} mutual impedance of line for the zero-sequence network

APPENDIX

Apparent Impedance for Three Phase Fault

The apparent impedance seen by a conventional reactance relay or fault locator will be derived for a 3-phase fault, including load current effects.

Load current flow in the line will be developed by phase angle a_E between two sources, per Fig. 2. Shunt capacitances and reactances will be neglected.

The derivation will use the superposition principle with Thevenin's theorem, where the total voltage and current at any point in the network is the sum of the prefault values plus those produced by short circuiting the voltage sources and inserting U_F, the open-circuit voltage at the fault point, as shown in Fig. 3. The polarity of U_F in the equivalent circuit of Fig. 3 is such as to cancel the pre-fault voltage except for the fault resistance drop. For the 3-phase fault case, the negative- and zero-sequence voltage drops are zero.

The positive direction of the load current I_{LD} is arbitrarily chosen as shown in Fig. 2.

The positive-sequence prefault voltage at station A is:

$$\begin{aligned}(U_{1A})_L &= U_{AE} - I_{LD} Z_{1SA} \\ &= U_F + I_{LD}\, p\, Z_{1L} \qquad \text{(A1)}\end{aligned}$$

The positive-sequence change in voltage at A produced by the fault is:

$$\begin{aligned}(U_{1A})_F &= - D_{1A} I_{1F} Z_{1SA} \\ &= D_{1A} I_{1F}\, p\, Z_{1L} + I_{1F} R_F - U_F \qquad \text{(A2)}\end{aligned}$$

The actual voltage at A is:

$$\begin{aligned}U_{1A} &= (U_{1A})_L + (U_{1A})_F \\ &= U_F + I_{LD}\, p\, Z_{1L} + D_{1A} I_{1F}\, p\, Z_{1L} \\ &\quad + I_{1F} R_F - U_F \\ &= I_{LD}\, p\, Z_{1L} + (D_{1A}\, p\, Z_{1L} + R_F) I_{1F} \qquad \text{(A3)}\end{aligned}$$

The actual current in the line at A is:

$$\begin{aligned}I_{1A} &= I_{LD} + \Delta I_{1A} \\ &= I_{LD} + D_{1A} I_{1F} \qquad \text{(A4)}\end{aligned}$$

From (A3) and (A4):

$$Z_A = U_{1A}/I_{1A}$$

$$= \frac{p\,Z_{1L}\,(I_{LD} + D_{1A}\,I_{1F}) + R_F\,I_{1F}}{I_{LD} + D_{1A}\,I_{1F}}$$

$$= p\,Z_{1L} + R_F\,\left(\frac{I_{1F}}{I_{LD} + D_{1A}\,I_{1F}}\right)$$

$$= p\,Z_{1L} + \frac{R_F}{D_{1A} + \frac{I_{LD}}{I_{1F}}} \qquad (A5)$$

From Fig. 3:

$$I_{1F} = \frac{U_F}{R_F + D_{1A}\,(p\,Z_{1L} + Z_{1SA})} \qquad (A6)$$

$$\text{where } D_{1A} = \frac{(1-p)\,Z_{1L} + Z_{1SB}}{Z_{1SA} + Z_{1L} + Z_{1SB}} \qquad (A7)$$

Eq. (A6) in (A5):

$$Z_A = p\,Z_{1L} + \frac{R_F}{D_{1A} + I_{LD}\,\frac{R_F + D_{1A}\,(p\,Z_{1L} + Z_{1SA})}{U_F}} \qquad (A8)$$

Eq. (A8) is a quadratic in p, the unknown fault location. R_F is also unknown.

Note that the argument for the R_F term is influenced by both the load current angle and by the impedances.

For a long line and a fault near station B, D_{1A} in Eq. (A7) approaches:

$$D_{1A} = \frac{Z_{1SB}}{Z_{1L}}$$

Assuming that Z_{1SB} has a larger argument than Z_{1L}, D_{1A} will have positive argument.

Neglecting load current, Eq. (A8) reduces to:

$$Z_A = p\,Z_{1L} + \frac{R_F}{D_{1A}}$$

This is plotted in Fig. A1 with the locus for varying R_F; the curve with the load effect included is also plotted.

The apparent-reactance effect can also be seen on a phasor diagram such as Fig. A2, with the current plotted horizontally as the reference.

Fig. A2 can be derived by rearranging Eq.(A3):

$$U_{1A} = p\,Z_{1L}\,(I_{LD} + D_A\,I_{1F}) + I_{1F}\,R_F$$

$$= p\,Z_{1L}\,(I_{1A}) + I_{1F}\,R_F \qquad (A9)$$

where I_{1A} = actual line current at station A.

Thus, a conventional reactance distance relay or fault locator using the same principle is subject to an error because of fault resistance, load current and differences in system-impedance arguments. The error shown in Figs. A1 and A2 is negative; however, for an import condition at station A (U_{AE} lags U_{BE}), the error will be positive.

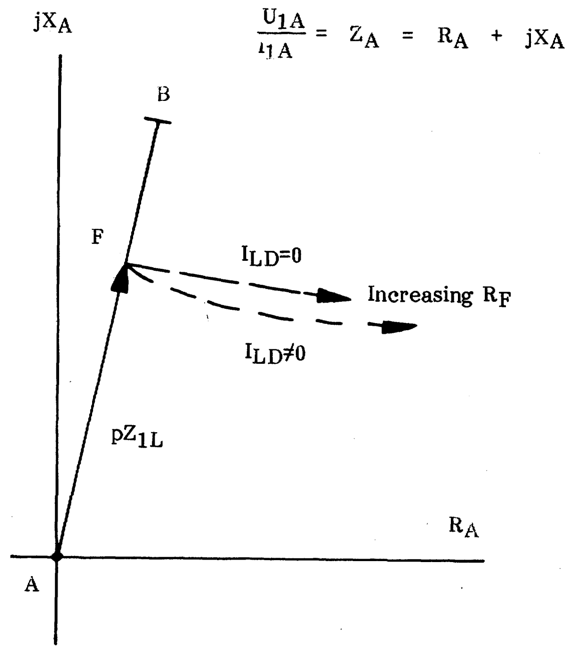


Fig. A1: Apparent impedance Z_A.

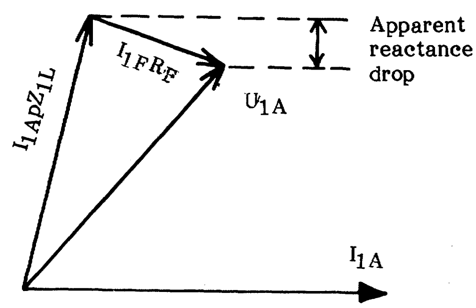


Fig. A2: Phasers showing apparent reactance effect in fault point potential.

Discussion

J. M. Crockett (Westinghouse Canada Inc., Hamilton, Ontario, Canada): The authors have outlined an interesting concept in fault locator design but some additional information would be useful.

The derivation of the equivalent source impedances Z1SA and Z1SN could be described in greater detail. It would appear that they are not actual source impedances but based on a system reduction to a two line machine equivalent and then the delta star conversion of the equivalent parallel line and source impedances. If the resulting mutual branch is eliminated, positive sequence source impedances result which yield correct distribution factors for all fault locations. Is this the approach used? It is not exact in the zero sequence network if mutual coupling is involved.

The positive sequence equivalent is not precisely rigorous if shunt loads at the line terminals are ignored or variable, and the degree of precision as regards fault locator accuracy may be very important even though the resulting source impedance argument variation is less than 3 degrees.

As an example of errors, consider source to line impedance ratio in the order of 0.5, shunt loads in the order of 12 or more times the source impedance, a substantial phase shift between sources, a high resistance fault to ground such as a tree, and a fault locator at the power receiving line terminal. I believe substantial changes in measured fault location would be observed as the shunt load, fault location, and fault impedance vary. Would the authors agree?

The ability to locate a high impedance fault to foliage would indeed be useful but has this capability been demonstated? Slow relaying would be expected for such a fault and hence apparent current and voltage deviations could be zero when the line was tripped.

Finally, I believe that the power-sending terminal is the preferred fault locator position. Would the authors agree?

Manuscript received August 2, 1984.

L. Eriksson, M. M. Saha, and **G. D. Rockefeller:** The authors thank Mr. Crockett for his interest in this paper and for his relevant comments and questions. His discussion can be segregated into four areas:

1. derivation of the current distribution factor for the parallel line case
2. effect of shunt loads
3. availability of prefault current information for a high-impedance fault
4. preferred spot for the locator.

These item numbers will be used below.

1. The current distribution factor as stated in Eq. 13A is rigorous assuming an identical parallel line. It is independent of any zero-sequence effect, including mutual impedance, because only the positive-sequence factor is needed to derive the total fault current I_f.

Measured voltage will contain any mutual induction from a parallel line—this can be compensated for by inputting the zero-sequence mutual impedance to the locator and providing the program with the measured parallel-line zero-sequence current via the printer loop, from the companion locator.

The Eq. 13A expression is derived using a delta-star conversion, where the source impedances are merged into the equivalent impedances. However, the *actual* source impedances are input just as for the single line algorithm which uses Eq. 13 for the distribution factor.

2. Shunt impedance effects at the two ends of the line can be rigously canceled by including these impedances in the source impedances. This is the case for either the single line or the parallel line algorithm. The load and source impedances are merely paralleled. Mr. Crockett is correct that changes in these, coupled with fault resistance, will introduce errors. The magnitude of the load impedance will have little effect, but its angular difference compared to the other impedances will have an effect as the impedance varies from the nominal value used to determine the source impedance input. The following example illustrates this effect.

A computer simulation on a 400 kV, 150 km line with a shunt load applied at Station A of the proportions stated by Mr. Crockett produced these results:

Case	Fault Point	R_f Ohms	Source Impedance Scalar	Angle	Error %
1	0.5	20	23	90	1
2	0.5	20	22.9	85.2	0
3	0.5	100	23	90	3
4	0.5	100	22.9	85.2	0
5	0.8	100	23	90	4
6	0.8	100	22.9	85.2	0

All six faults were phase to ground. Cases 2, 4 and 6 have zero error, because the load impedance has been rigorously accommodated. Cases 3 and 5 represent extreme conditions, with 100 ohms of fault resistance, particularly case 5 where the fault is near the far end. By choosing a representative value for the load impedances variations in this value will have a limited error effect.

The parameters for this example are:

	magnitude	angle-deg.
Z_{1SA}	23	90
Z_{0SA}	23	90
Z_{SH}	276	0
Z_{1SB}	20	90
Z_{0SB}	40	90
Z_{1L}	46.5	84.4
Z_{0L}	156.7	77.5
U_{AE}	230.9	0
Z_{BE}	230.9	-38.5
I_{LD}	2312	-14.3
Z_{SP}	22.9	85.2

Impedances are in ohms, voltages in kV, currents in amperes and angles in degrees. Z_{SF} is the positive-sequence equivalent of the paralleled source Z_{SA} and load impedance Z_{SH}. Note that the effect of the load impedance at Station A is to shift the actual source angle from 90 deg. to 85.2 deg. for the paralleled equivalent of the source and load impedances. The load has a negligible effect on the magnitude of this equivalent.

3. If the normal high-speed proteciton fails to detect a high-resistance fault, such as a mid-span flashover to foliage, two possibilities exist for the locator:

a) The fault-locator terminal opens following the remote terminal. In this case, there is no load current flow during the fault period used by the locator to compute. Thus, no error occurs because of load current.

b) The fault-locator terminal opens first. The prefault data are not available in memory if the line protection trip is delayed. This situation will be recognized by the program and an alternative "slow trip" routine is processed. Errors will occur because of the load current component.

4. The direction of power flow is not of concern to the fault locator, since its algorithm eliminates load flow effects. That is, there is no bias towards installing the locator at the "power sending terminal" unless this describes the *stronger* source of fault current. The stronger source installation reduces errors because the weaker end produces less of an infeed effect.

5. The following additional field results are noteworthy. Six phase-ground faults occurred during very high winds on a 400 kV, 135 km line. The precise location of these temporary faults is not known. The locator operated in each case, indicating in the range of 93 to 99%.

Manuscript received September 4, 1984.